Chapter 18
Indigenous Research and Data Management in Electronic Archives:
A Framework for African Indigenous Communities

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ABSTRACT

This chapter introduces a framework for indigenous research and data management in electronic archives that aligns with indigenous worldviews and practices. It discusses indigenous communities' challenges in owning and controlling their data and the need for a culturally relevant framework for managing indigenous data in electronic archives. The proposed eight-step framework emphasises community control, data sovereignty, and ethical data management practices; and includes key components such as community engagement, informed consent, and culturally relevant metadata standards. Best practices for data sharing and partnership building with non-indigenous institutions are also discussed, as well as the steps for implementing the framework and the role of stakeholders in the process. Evaluation metrics for measuring the framework's success are proposed. The chapter concludes by emphasising the importance of community control and ethical data management practices in preserving and protecting indigenous cultural heritage and identity in electronic archives.

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INTRODUCTION

Indigenous data management in electronic archives has significant implications for indigenous communities, research, and society (Smith, 2021). It is crucial to empower indigenous communities to take ownership and control of their data, ensuring that their cultural heritage and knowledge are preserved and protected for future generations (Kim et al., 2019; Ruhanen & Whitford, 2019). This promotes self-determination and reinforces the value of indigenous knowledge systems and practices. In addition, it enables researchers to conduct more culturally sensitive and ethical research by engaging with indigenous communities and gaining a deeper understanding of their knowledge, culture, and values (Anderson & Cidro, 2019; Hyett et al., 2019; Mashford-Pringle & Pavagadhi, 2020). Furthermore, indigenous data management in electronic archives promotes knowledge sharing and collaboration between indigenous and non-indigenous communities, contributing to recognising and appreciating indigenous knowledge and perspectives. This can help bridge the gap between indigenous and non-indigenous communities and promote social cohesion (Gupta et al., 2023). By acknowledging the importance of indigenous data and collaborating with indigenous communities, non-indigenous institutions can help to rebuild trust and support broader societal goals of reconciliation and social justice (McGregor et al., 2023).

A framework for indigenous research and data management in electronic archives is therefore, needed to be guided by indigenous worldviews and practices. This chapter proposes an 8-step framework that includes indigenous data collection, curation/processing, analysis/synthesis, storage, security/backup, retrieval, sharing/communication, and reuse. The framework is guided by community control, data sovereignty, and ethical data management practices that align with indigenous worldviews and practices. The chapter thus discusses best practices for indigenous data management in electronic archives, emphasising the importance of developing data-sharing protocols and establishing partnerships between indigenous communities and non-indigenous institutions.

CONCEPT OF INDIGENOUS RESEARCH AND DATA

Indigenous research is an approach that values and respects indigenous knowledge, perspectives, and ways of knowing. It involves collaborating with indigenous peoples to promote self-determination and empower indigenous communities. Smith (2021) argues that a critical examination of research practices and the impact of colonialism on indigenous peoples is necessary for indigenous research. This approach recognises the importance of oral traditions, community-based knowledge and the use of indigenous languages in research (Wilson, 2020). In addition, indigenous research requires a culturally sensitive approach that respects indigenous protocols and values. This includes obtaining free, prior and informed consent from indigenous communities before conducting research while also ensuring that research benefits the community, and protecting the privacy and confidentiality of research participants (Bassey & Owan, 2023; Owan et al., 2023). Smith (2021) emphasises that a decolonising approach to research must challenge the dominant research paradigm and promote indigenous knowledge and ways of knowing.

On the other hand, indigenous data refers to the information generated, collected and analysed by indigenous communities for their purposes and decision-making processes. This data is based on indigenous knowledge systems, cultural practices and ways of knowing and often includes traditional ecological knowledge, oral histories, and other forms of knowledge that are not typically captured in

Western-style data collection methods (Smith, 2021). Indigenous data has gained increasing recognition in recent years as a way to empower indigenous communities to take ownership of their data and use it for their purposes. This has been partly driven by concerns about the exploitation of indigenous data by non-indigenous researchers and institutions and the need to support indigenous self-determination and governance (Oguamanam, 2020). Indigenous data sovereignty is a related concept that refers to the right of indigenous peoples to control the collection, ownership, and use of their data. This includes the right to determine how data is collected and who has access to it, as well as the right to use data for their purposes and to protect it from exploitation (Kukutai & Taylor, 2016).

TYPES OF INDIGENOUS DATA

There are several types of indigenous data which include:

- **Traditional Knowledge**

This type of data encompasses the knowledge, practices, and beliefs passed down through generations of indigenous peoples. It often includes environmental knowledge, such as traditional land management practices, medicinal plants, and wildlife behaviour (Walter & Suina, 2019). It can also include cultural knowledge, ceremonial practices, language and art. Traditional knowledge is often based on a deep understanding of the natural world and reflects indigenous perspectives and values.

- **Oral Histories**

Oral histories are a type of indigenous data that are transmitted through verbal narratives and storytelling from one generation to the next. This type of data often includes stories, legends, and myths that explain the origins of the world and the values and beliefs of indigenous communities. Oral histories can provide insight into the historical and cultural contexts in which indigenous communities exist and can serve as a way to preserve cultural knowledge.

- **Indigenous Language Data**

Indigenous language data refers to indigenous languages' words, grammar, and syntax. This data type is often used to understand indigenous peoples' linguistic and cultural heritage. Many indigenous languages are endangered or have been lost, so the preservation and study of language data are crucial for the continuation of indigenous cultures (Ruckstuhl, 2022).

- **Community-Based Research Data**

Community-based research data is produced through research projects collaborating with indigenous communities. This type of data often reflects the perspectives and priorities of indigenous peoples and can be used to address community needs and concerns, and it is often guided by reciprocity, respect, and shared decision-making principles (Arxer, 2019; Jason & Glenwick, 2016).

- **Genealogical Data**

Genealogical data is a type of indigenous data that relates to family histories and lineages. This type of data is often used to understand indigenous communities' social and cultural structures, kinship systems, and inheritance practices. Genealogical data can establish legal and cultural connections to traditional lands and territories (Gavrilov et al., 2002; Liu et al., 2017).

- **Environmental Data**

Environmental data refers to indigenous peoples' observations and knowledge about their natural environment. This type of data is often used to inform environmental management and conservation efforts. indigenous communities have a long history of living in a close relationship with the natural world, and their knowledge can be invaluable for understanding ecological systems and developing sustainable practices.

- **Artistic and Cultural Data**

Artistic and cultural data includes a range of creative expressions, such as music, dance, and visual arts. This type of data often reflects indigenous communities' cultural traditions and values. It can provide insight into indigenous peoples' histories, beliefs, and practices and serve as a means of cultural expression and preservation.

CONCEPT OF ELECTRONIC ARCHIVES

Electronic archives refer to collections of digital records that are preserved and made available for future access and use (Owan, 2022). The International Council on Archives described electronic archives as a set of digital records, preserved for their long-term value, authenticity and reliability, and made accessible to users over time (ICA, 2017). Electronic archives can include various types of digital content such as text documents, images, audio and video recordings and data sets. Electronic archives are becoming increasingly important as more information is being created and stored in digital format, and as the need for long-term preservation of this information becomes more pressing. However, the electronic archives must be authentic, reliable, usable and have the integrity to be valuable to society. Thus, electronic archives are typically managed using specialised software to ensure the integrity and authenticity of digital records and provide search, retrieval and access control mechanisms.

Types of Electronic Archives

The following are types of electronic archives:

- **Digital Document Archives**

Digital document archives store digital versions of physical documents such as books, manuscripts, photographs, maps, and other primary source materials. They are often used by scholars and researchers studying history, literature, and other humanities subjects. These archives allow users to access primary source materials from anywhere worldwide, which can be especially useful for researchers who cannot visit physical archives. Digital document archives also help preserve physical materials that may be fragile or deteriorating, ensuring they are available for future generations (Matlala, 2019).

- **Web Archives**

Web archives capture and preserve web pages and other online content, providing a snapshot of how the internet looked at a particular point in time, and are created by web crawlers, which automatically scan and capture web pages and other online content (Vlassenroot et al., 2019). These archives are important for researchers studying social, cultural and political phenomena documented online, such as the development of social media or the evolution of online activism.

- **Multimedia Archives**

Multimedia archives store audio and video recordings, photographs, and other digital media. Multimedia archives are used in various fields, including journalism, film studies, and anthropology. These archives may be organised by subject, date or other criteria and often include metadata such as descriptions and tags that make it easier to find specific media files. Multimedia archives also play an important role in preserving cultural heritage, as they allow for the preservation and sharing of traditional music, dance, and other forms of artistic expression.

- **Social Media Archives**

Social media archives are typically created by web crawlers or other automated tools that capture content from social media platforms such as Twitter, Facebook, and Instagram (Anderson, 2020; Pehlivan et al., 2021). These archives capture and preserve social media content, including posts, comments and other interactions, and are used by researchers studying topics such as social movements, political campaigns, public opinion as well as journalists and other media professionals to track breaking news and monitor public sentiment.

- **Research Data Archives**

These archives store data from scientific studies and other research projects. Scientists and researchers in various fields, including biology, physics, and social sciences, use research data archives. They may be organised by discipline or topic, often including metadata describing the data and its sources. Research data archives help to ensure that scientific data is available for future studies and can be used to verify or build upon existing research (Bossaller & Million, 2022).

- **Institutional Archives**

Institutional archives store digital records and documents related to the activities of an institution such as a university or government agency, and are often used for administrative purposes such as tracking budgets, personnel records and other institutional data. These archives may also include reports, policy documents and other materials documenting the institution's activities.

- **Personal Archives**

Personal archives store personal digital records such as email correspondence, photographs and personal documents. Individuals or families may create and maintain personal archives to preserve history and memories and may include family photographs, personal journals and other materials documenting the individual's or family's life and experiences.

ELECTRONIC ARCHIVES AND INDIGENOUS DATA MANAGEMENT: THE NEXUS

Electronic archives and indigenous data management are two concepts that have become increasingly important in recent years. The nexus between these two concepts lies in ensuring that indigenous data is managed in a culturally appropriate manner and that electronic archives do not further perpetuate historical injustices. Indigenous data management is a critical issue because data collected from indigenous communities have historically been used to dispossess them of their land, culture, and rights (Cormack & Kukutai, 2022). The collection of this data has often been done without the informed consent of the communities, and the data has been used to justify discriminatory policies and practices. As a result, indigenous communities have become increasingly distrustful of external researchers and institutions that collect and use their data (Figueiredo et al., 2020). The field of indigenous data sovereignty has emerged in response to these issues (Rainie et al., 2019; Walter et al., 2021). This field recognises that indigenous communities have the right to own, control and govern their data. Indigenous data sovereignty involves working with indigenous communities to ensure that their data is collected, managed and analysed culturally appropriately. This approach requires a recognition of the importance of indigenous knowledge systems and a commitment to building relationships of trust and respect with indigenous communities.

Electronic archives can play an important role in supporting indigenous data sovereignty. The digitisation of information can make it more accessible to indigenous communities and help to preserve their cultural heritage. However, the digitisation of information can also be a source of concern if it is done without the appropriate cultural protocols and consent from indigenous communities. Electronic archives must be designed and implemented to respect indigenous data sovereignty and support community goals. One way to ensure electronic archives are designed and implemented culturally is to involve indigenous communities. Indigenous communities should be involved in planning, designing, and implementing electronic archives that relate to their data, and this involvement should be based on the principles of informed consent and respect for cultural protocols. Indigenous communities should also be consulted on issues such as access, storage, and use of their data in electronic archives. These communities should also be involved in decisions about using digital technologies and tools to manage, analyse and preserve their data.

Another important issue related to electronic archives and indigenous data management is the protection of privacy and confidentiality. Indigenous communities can control their data and decide who can access it. Electronic archives must thus be designed with appropriate security measures to protect the privacy and confidentiality of indigenous data and this includes using encryption and access controls to prevent unauthorised access and ensuring that data is only used for purposes agreed upon by the community.

IMPORTANCE OF INDIGENOUS DATA MANAGEMENT IN ELECTRONIC ARCHIVES

The following are some reasons why managing indigenous data in electronic archives is important:

- **Complete Data Security**

Regarding indigenous data and knowledge, complete data security is particularly important as this information may be sensitive and sacred to the community. Storing this information in physical files can be risky as they may be lost, damaged or accessed by unauthorised persons. Digital archival provides a secure way of storing indigenous data and knowledge, ensuring that it is protected and only accessible to authorised individuals (Hunter, 2005). With digital archival, you can encrypt and password-protect files, use firewalls and antivirus software to protect against cyber threats, and backup files to prevent data loss (Maina, 2012). This level of security is crucial for protecting the integrity of indigenous data and knowledge and ensuring its long-term preservation.

- **Simpler to Manage**

The digital archival of indigenous data or knowledge in digital archives allows for easier sharing and collaboration between individuals and organisations. Physical documents may be limited to one location, making it difficult for individuals in different locations to access the same information. With digital archives, access to data is not limited by location, and multiple users can access the same data simultaneously, regardless of location (Müngen, 2022). This makes it easier for indigenous communities to collaborate and share their knowledge with others who may be interested in learning from them. In addition, the ease of access and sharing allows for better protection of indigenous data or knowledge from unauthorised use, as access can be controlled and monitored. Thus, digital archival simplifies managing and sharing indigenous data/knowledge while providing better protection against misuse.

- **Cost Saving**

Digital archival can also result in significant cost savings (Boczar et al., 2023; Jahn et al., 2020; Pillen & Eckard, 2023). Indigenous communities may have limited resources and budget constraints, which makes it challenging to maintain and preserve physical documents. By switching to digital archival, indigenous communities can reduce inventory and stationery costs and save on the expenses of transporting physical documents to different locations. This is especially important for preserving indigenous knowledge across different regions or communities. Moreover, digital archival can also help reduce the costs of managing and preserving physical documents. Indigenous communities can easily organise, store, and retrieve digital documents without worrying about physical storage limitations or damage due to environmental factors. This makes it easier for indigenous communities to access and share their knowledge, supporting their cultural preservation and development efforts.

- **Collaborative Access and Editing**

Digital archives make it easier for multiple users to access and edit the same document simultaneously (Goldin et al., 2022), even in different locations (Cannelli & Musso, 2022; Jaillant, 2022; Jaillant & Rees, 2023). This allows for more efficient collaboration among individuals and teams working on a project. For example, multiple indigenous communities and organisations can collaborate on a digital archive to share their knowledge and preserve their cultural heritage.

- **Improved Search and Retrieval**

Digital archives use advanced search algorithms, making locating specific documents or data easier. Users can search for documents using keywords, tags, or other metadata (Owan, 2022), and this makes it easier to find relevant information quickly and efficiently.

OVERVIEW OF EXISTING FRAMEWORKS AND MODELS FOR INDIGENOUS DATA MANAGEMENT

Indigenous data management is a complex and evolving field that requires consideration of cultural, ethical and legal factors. Recently, there has been an increased focus on developing culturally appropriate, respectful, and effective frameworks and models for indigenous data management. This chapter reviewed existing frameworks and models for indigenous data management and discuss their key features and limitations.

The National Aboriginal Health Organization (NAHO) in Canada developed one of the earliest frameworks for indigenous data management. The framework is known as the Ownership, Control, Access, and Possession (OCAP) principles that guides indigenous communities and organisations in managing their data in a culturally appropriate and respectful manner (Kukutai & Taylor, 2016). The OCAP principles emphasise the importance of respecting indigenous communities' ownership and control of data, ensuring that data is accessible to community members, and protecting the confidentiality of personal and sensitive data (First Nations Information Governance Centre, 2014).

Another key framework for indigenous data management is the Te Mana Raraunga framework developed by the Maori Data Sovereignty Network in New Zealand (Maori Data Sovereignty Network, 2019). This framework is based on the principles of mana (authority), tiaki (guardianship) and kaitiakitanga (stewardship) and emphasises the importance of respecting Maori cultural values and practices in data management. The framework guides on issues such as data governance, data ownership, and data sharing and emphasises the importance of community involvement in decision-making processes (Maori Data Sovereignty Network, 2019).

In addition to these frameworks, several models for indigenous data management have been developed in specific contexts. For example, Canada's Inuvialuit Settlement Region (ISR) has developed a data management model emphasising community involvement in data collection, analysis and interpretation (Inuvialuit Regional Corporation, 2017). The model involves establishing a community-based data management system controlled by the community and providing access to data relevant to the community's needs and interests.

Another model for indigenous data management is the Indigenous Data Sovereignty model developed by the Aboriginal and Torres Strait Islander Data Archive in Australia (Aboriginal and Torres Strait Islander Data Archive, 2021). This model emphasises the importance of indigenous self-determination in data management and calls for developing indigenous-led data governance structures grounded in indigenous cultural values and practices.

Despite the growing number of frameworks and models for indigenous data management, some key challenges and limitations still need to be addressed. One of the main challenges is the lack of resources and capacity for indigenous communities and organisations to implement these frameworks and models effectively (Kukutai & Taylor, 2016). This can result in a lack of standardisation and consistency in data management practices, limiting the usefulness and comparability of data across different contexts. Another challenge is balancing the competing demands of data protection and sharing (First Nations Information Governance Centre, 2014). Indigenous communities are often concerned about using their data by external researchers or government agencies and may be hesitant to share data outside the community. However, data sharing is also important for promoting collaboration and knowledge exchange and can help to address health and social disparities. Another challenge is that indigenous data management's ethical and legal considerations constantly evolve and newer frameworks and models are needed to reflect these changes. Furthermore, the impact of globalisation and digitalisation on indigenous communities is complex and multifaceted, and newer frameworks and models are needed to address the challenges and opportunities that arise from these trends.

A FRAMEWORK FOR INDIGENOUS RESEARCH AND DATA MANAGEMENT

The framework developed in this chapter has eight broad steps, including indigenous data collection, indigenous data curation/processing, indigenous data analysis/synthesis, indigenous data storage, indigenous data security/backup, indigenous data retrieval, indigenous data sharing/communication, and indigenous data reuse. These steps were extracted from the 16 data management practices in previous work (Owan & Bassey, 2019), and the five data management practices in another (Odigwe et al., 2020). The eight steps were selected to develop the framework in this chapter carefully due to their applicability in indigenous data management systems.

Step 1: Indigenous Data Collection

Indigenous data collection refers to the process of collecting and analysing data that is relevant to indigenous communities, and this data can be used for various purposes, including improving policies, programs, and services that affect indigenous peoples. Indigenous data collection typically involves working with indigenous communities to understand their specific needs and priorities and ensuring that data is collected in a culturally appropriate and respectful manner. This may involve developing partnerships with indigenous organisations, using indigenous research methodologies, and involving indigenous community members in all stages of the data collection process. Generating data requires appropriate methods and instruments to ensure the validity of study findings (Sileyew, 2020). It is crucial to collect data accurately, as erroneous data can lead to misleading research results (Owan & Bassey, 2019). To collect accurate data, researchers and their teams must have the necessary skills and knowledge, while respondents or the studied phenomena must provide truthful details. The framework to guide indigenous data collection is presented in Table 1.

Step 2: Indigenous Data Curation/Processing

Data curation refers to the process of selecting, organising, and maintaining data for use in research, analysis, or other purposes. It involves ensuring that data is consistent, accurate, and high-quality. Data curation involves several steps, including cleaning, normalisation, and integration. The data collection process may result in not immediately usable data, requiring data cleaning to improve data quality (Owan & Bassey, 2019). Subsequently, meaningful information can be derived through data analysis, leading to potential problem resolution or improvement (Banerjee et al., 2013). On the other hand, data processing is manipulating, transforming, and analysing data to derive insights or create useful information. Data processing involves several steps, including transformation and visualisation (French, 1996). Table 2 presents a framework that can be used to direct the curation and processing of indigenous data.

Step 3: Indigenous Data Analysis/Synthesis

Data analysis is organising and interpreting data to extract valuable insights or information. It involves breaking down the data into its components or categories better to understand each (Owan & Bassey, 2019). Soler et al. (2016) describe data synthesis as combining diverse data sets from various sources, such as incorporating newly collected data with pre-existing data from different providers, to create a larger and more comprehensive dataset. Therefore, indigenous data analysis/synthesis refers to analysing and synthesising data collected from indigenous communities or relevant to indigenous peoples' experiences and perspectives. The goal is to gain a more comprehensive understanding of the studied issues and ensure that the findings contribute to positive outcomes for indigenous communities. A framework is provided in Table 3 that can guide the analysis and synthesis of indigenous data.

Table 1. A framework for indigenous data collection

Stakeholders	Activity/Expectation
Research team	The research team establish protocols and guidelines for data collection that prioritise the community's cultural traditions and knowledge systems. They work with community elders and knowledge keepers to collect data using culturally appropriate methods.
Indigenous community	The indigenous community is at the centre of this framework as their cultural traditions and knowledge systems are prioritised in the data collection process. They guide the research team in collecting, storing, and sharing data. The community also has the power to decide whether to participate in the research, and their consent must be obtained before any data collection can occur.
Elders and knowledge keepers	Elders and knowledge keepers are crucial stakeholders in this framework as they hold the community's cultural knowledge and traditions. They provide the research team with all relevant information through storytelling or responding to questions seeking further insights into their history, activities and practices.
Government agencies	Government agencies play a role in this framework by providing funding and support for the research project. They also ensure that the project follows ethical guidelines and standards for data collection, storage, and sharing. Government agencies are expected to perform supervisory and oversight functions.
Academia	Academia provides expertise and knowledge in data collection, analysis, and interpretation. Academia is expected to provide technical support to the research team, such as instrument development and validation, ensuring adherence to ethical standards.

Table 2. A framework for indigenous data curation and processing

Stakeholders	Activity/Expectation
Research team	The research team plays a critical role in this stage, as they are responsible for organising and processing the data using indigenous data curation tools and techniques. They must have expertise in indigenous data curation and be sensitive to the cultural context of the data.
Indigenous community	The indigenous community has a stake in curating and processing the data as it is their knowledge being represented. Community members can help identify the data types that are most important and relevant to their community. This can involve specifying the topics, issues, and questions the data should address.
Data managers	Data managers can work with indigenous communities to develop data management plans that outline how the data will be collected, organised, stored, and shared. Data managers can help ensure the quality and integrity of indigenous data by implementing quality control measures, such as data cleaning and validation procedures. They must have expertise in data management and indigenous data curation to ensure that the data is treated with respect and sensitivity.
Technology providers	Technology providers may be involved in this stage as they provide tools and platforms for the organisation and processing of the data. They must ensure their tools are culturally appropriate and respect indigenous knowledge systems.
Funding agencies	Funding agencies play a role in this stage as they support the curation and processing of the data. They may require that the data is organised and processed in a way consistent with best practices for indigenous data curation.

Table 3. A framework for indigenous data analysis and synthesis

Stakeholders	Activity/Expectation
Research team	The research analyses data using methods appropriate for indigenous research, such as community-based participatory research (CBPR) or two-eyed seeing. They also synthesise the data collected from indigenous communities to generate insights that are relevant and meaningful to them. This may involve identifying patterns, themes, or trends in the data and interpreting these findings in the context of indigenous knowledge systems and experiences.
Indigenous community	The indigenous community ensure that their perspectives are represented in the research. This can include participating in focus groups, workshops, or interviews to help interpret the data. The indigenous community validates the findings to ensure that the insights generated are accurate and meaningful. This process can involve reviewing reports or other materials and providing feedback to the research team.
Elders and knowledge keepers	The elders and knowledge keepers provide a cultural and spiritual lens. They ensure that the data is analysed respectfully and ethically and that the findings are interpreted within the community's cultural context. Their insights can also help to identify gaps in the data or areas where further research is needed. Elders and knowledge keepers provide valuable guidance in interpreting the data, particularly about traditional knowledge and practices. They contextualise the findings and ensure they are interpreted consistently with the community's values and beliefs.
Academia	In this stage, Academia provides technical expertise in data analysis and synthesis. This may involve using specialised software and statistical techniques to analyse large datasets or developing novel methods for analysing qualitative data. Academics also collaborate with elders and knowledge keepers to incorporate their insights and perspectives into the analysis process and ensure that the findings are presented in a culturally appropriate and respectful manner. Academics bridge the gap between indigenous communities and mainstream scientific communities by presenting research findings in a way that is accessible to both groups. This may involve using plain language summaries or visualisations to help communicate complex data to a broader audience.

Step 4: Indigenous Data Storage

Indigenous data storage refers to the various methods and systems used to store, manage, and protect data collected from or relevant to indigenous communities. Storing research data for future reference is a crucial process that enables researchers to retrace their steps in case of errors in analysis (Owan & Bassey, 2019). This includes traditional and modern storage methods like paper records, digital files, and cloud-based solutions. Data sovereignty is a key concept in indigenous data storage (Hummel et al., 2021; Lovett et al., 2019), which recognises the right of indigenous peoples to control and manage their data. However, the framework developed in Table 4 provides a guide for storing indigenous data in electronic archives.

Step 5: Indigenous Data Security/Backup

Data security involves implementing protective measures to prevent unauthorised access to private data across various platforms, such as computer systems, databases, websites, and mobile devices (Owan & Bassey, 2019). To ensure data security, it is necessary to consider physical security, network security, and the security of computer systems and files to prevent unauthorised access, modification, disclosure,

Table 4. A framework for indigenous data storage in electronic archives

Stakeholders	Activity/Expectation
Research team	The research team stores indigenous data in electronic archives after obtaining informed consent from participants. They ensure that the data are collected, managed, and stored culturally appropriately. The research team provides appropriate safeguards to protect participants' privacy and confidentiality. The research team collaborates with indigenous communities to ensure that the data is being stored and managed in a way that aligns with the values and beliefs of the community. This may involve engaging with elders and knowledge keepers to gain insights into traditional knowledge systems and cultural protocols.
Indigenous community	The indigenous community must be involved in the decision-making process around the storage and use of the data, and their informed consent should be obtained before any data is collected or stored. The community works with the research team to establish policies and practices that align with the community's values and beliefs and develop protocols for accessing and using the data. They should have access to information about the data storage and management processes and be informed about any potential risks or benefits associated with using their data. The community should also be involved in the ongoing management and governance of the data, including decisions around access and use. They should have a say in how the data is used and can revoke consent or limit access to the data if they feel that their privacy or cultural protocols are being compromised.
Data managers	Data managers ensure that the data is stored securely, with appropriate measures to prevent unauthorised access, loss, or theft. This may include implementing encryption and access control measures, regular backups, and disaster recovery planning. Data managers must document the data to make it easy to understand and use. This may include providing detailed metadata, documentation of the data collection process, and information on any relevant cultural protocols or ethical considerations. Data managers must ensure that the data is accessible to authorised users while also taking steps to protect the privacy and confidentiality of the data. This may involve implementing access controls, requiring users to sign confidentiality agreements, or providing remote access to the data through secure networks. Data managers must maintain the data over time, with appropriate measures to ensure its long-term preservation and accessibility. This may include regular data integrity checks, migration to new storage systems, and monitoring ongoing data usage and access.
Technology providers	Technology providers ensure that the electronic archives they provide are secure and confidential. This means implementing access controls, encryption, and firewalls to prevent unauthorised access, data breaches, and cyber-attacks. Technology providers ensure that the electronic archives are reliable, and the data is not lost or corrupted. This means providing redundant storage, backups, and disaster recovery plans to ensure that data can be recovered in case of a system failure or other disaster. Technology providers ensure that the electronic archives support appropriate data formats relevant to the indigenous communities' specific needs. This may include supporting languages, data structures, and data types relevant to indigenous communities. Technology providers must ensure that the electronic archives are accessible to the indigenous communities and that they can easily access and retrieve their data. Technology providers must ensure that the electronic archives are sustainable over time and can support the long-term storage and retrieval of indigenous data. This may include ongoing support, maintenance, and updates to ensure the archives remain relevant and useful to the indigenous communities.
Funding agencies	Funding agencies provide adequate funding to ensure indigenous data is properly stored and managed in electronic archives. This will enable researchers to access the data for future research purposes.

or destruction of data (Parker, 2012). Data backup enables researchers to safeguard research data against loss (Abduldayan et al., 2021). This involves storing the data across multiple files, locations, and media to ensure its availability during a disaster. Backup can be done manually using physical storage options or electronically through various online storage systems such as email, Google Drive, and cloud storage services (Owan & Bassey, 2019). In Table 5, a framework offering guidance on the security and backup of indigenous data in electronic archives has been developed.

Step 6: Indigenous Data Retrieval

Data retrieval allows authorised individuals to regain access to previously stored data, which can be used for various purposes such as verification, data reuse, error checking, observation, and reference (Owan

Table 5. A framework for indigenous data security and backup in electronic archives

Stakeholders	Activity/Expectation
Research team	The research team classify the indigenous data and ensure that access to it is restricted to authorised individuals only. They also implement access control measures such as password protection, encryption, and two-factor authentication to prevent unauthorised data access. The research team ensure that indigenous data are regularly backed up to prevent loss in case of a disaster. The research team ensures the electronic archives are regularly updated and upgraded to prevent vulnerabilities and ensure compatibility with new technologies. The team also ensures that any software or hardware used in the archives is patched and updated.
Indigenous community	The indigenous community asserts its ownership and control over the indigenous data ensuring that the data are being used in a manner that respects its cultural and intellectual property rights. The indigenous community gives informed consent and is consulted about using and sharing the indigenous data. The community should know who can access the data and for what purposes. The indigenous community should be vigilant in protecting the indigenous data from unauthorised access, use, and sharing. The indigenous community educate and raise awareness about the importance of indigenous data and its protection. The community should inform its members about the rights and responsibilities of using and sharing data.
Data managers	Data managers design and implement appropriate security measures to protect indigenous data. These measures may include access controls, firewalls, encryption, and intrusion detection systems. Data managers establish and implement regular backup and disaster recovery procedures to ensure indigenous data is protected and recoverable during a disaster or system failure. Data managers ensure that access to indigenous data is limited to authorised individuals only. The data storage infrastructure should be robust, reliable, and scalable to accommodate the growth of indigenous data. Access to the data should be controlled through authentication mechanisms, such as usernames and passwords. Data managers should provide training and awareness programs for personnel managing and using indigenous data.
Technology providers	Technology providers design and implement secure systems to protect indigenous data. The systems should use the latest security technologies and best practices to safeguard the data from unauthorised access, use, and disclosure. The systems should be scalable and flexible enough to accommodate indigenous data's growth and changing needs. Technology providers provide training and awareness programs for data managers and other personnel managing and using indigenous data.
Funding agencies	Funding agencies provide sufficient resources to support the development and maintenance of electronic archives. This includes the hardware, software, and staffing cost necessary to ensure the security and backup of indigenous data. Funding agencies develop clear policies and guidelines for the ethical use of indigenous data. Funding agencies ensure backup copies of indigenous data are created and stored in their directories or repositories in multiple locations to protect against data loss due to hardware failures, natural disasters, or other events.
Data Users	Data users comply with the security policies established by the funding agencies or institutions responsible for the electronic archives. Data users must understand the sensitivity of the data they are working with and treat it accordingly. Indigenous data may contain sensitive information about cultural practices, beliefs, and identities. Therefore, data users should be careful to protect this data from unauthorised access, loss, or theft. Data users should promptly report security incidents or breaches to the appropriate authorities. This can help minimise the incident's impact and prevent future incidents from occurring.

& Bassey, 2019). Indigenous data retrieval refers to accessing, recovering, and utilising data collected, generated, or preserved by indigenous communities. The retrieval of indigenous data can serve several purposes, including the preservation of cultural heritage, the advancement of indigenous research and education, and the empowerment of indigenous communities. Principles of cultural respect, ethical considerations, and community engagement guide the process of indigenous data retrieval. Table 6 presents a framework that can assist in retrieving indigenous information from electronic archives.

Step 7: Indigenous Data Sharing/Communication

Sharing data benefits individual researchers and the scientific community (Longo & Drazen, 2016; Tenopir et al., 2011). According to Soler et al. (2016), data can be shared within the research team with external collaborators and can also be made publicly available. Indigenous data sharing and communication refer to the exchange of information between indigenous communities and individuals and other stakeholders such as researchers, policymakers, and governments. It involves sharing data, knowledge, and information collected, produced, or held by indigenous peoples and communities. Table 7 provides a framework for indigenous data sharing and communication in electronic archives.

Step 8: Indigenous Data Reuse

Data reuse refers to using previously collected data for different purposes, such as validating prior research findings or addressing new problems (Owan & Bassey, 2019). Researchers often reuse data from public repositories, especially for similar studies conducted in different locations (Park & Wolfram, 2017).

Table 6. A framework for indigenous data retrieval from electronic archives

Stakeholders	Activity/Expectation
Research team	The research team identifies the electronic archives that contain relevant indigenous information. This may involve searching online databases, archives, libraries, and other repositories. The research team determine if the archives have access restrictions or require permissions to retrieve the information. If access or permissions are required, the team must obtain them before proceeding. The research team must review the archives and identify the specific indigenous information relevant to their research objectives. This may involve searching for specific keywords, topics, or themes. Depending on the type of information and the archive, the research team may need to use different retrieval techniques. The research team ensures they follow the appropriate cultural protocols when retrieving indigenous information. The research team must document the retrieval process to ensure transparency and reproducibility. This may include documenting the search terms, the date and time of retrieval, the sources consulted, and any issues encountered. Once the indigenous information has been retrieved, the research team must analyse and interpret it in the appropriate cultural and historical contexts. This may involve working with community members and cultural authorities to ensure that the analysis and interpretation are respectful and accurate.
Indigenous community	Indigenous communities can start by identifying the electronic archives that contain their relevant indigenous information. Indigenous communities can establish partnerships and collaborations with research institutions, archives, and other organisations to retrieve their indigenous information. Indigenous communities can use appropriate legal and ethical frameworks to protect their rights and interests in retrieving their indigenous information. Indigenous communities can use appropriate retrieval techniques to access their indigenous information from electronic archives. This may involve working with research institutions and archives to use appropriate search terms, metadata, and other techniques to retrieve relevant information. Indigenous communities can document the retrieval process to ensure transparency and reproducibility. This may include documenting the search terms, the date and time of retrieval, the sources consulted, and any issues encountered.
Data managers	If data managers, technology providers, funding agencies and other users are interested in retrieving indigenous data from electronic archives, then the process is the same as for the research team above.

Table 7. A framework for indigenous data sharing and communication in electronic archives

Stakeholders	Activity/Expectation
Research team	The research team must obtain informed consent from the indigenous communities before sharing or communicating their data. The research team must acknowledge the source of the data and give credit to the indigenous communities for their contributions. This includes appropriate attribution in any publications, reports, or presentations using the data.
Indigenous community	The indigenous community provides informed consent for sharing and communicating their data. The indigenous community review and approve any data-sharing agreements before they are finalised. The indigenous community provide feedback and guidance on the use of the data. This includes providing input on any publications, reports, or presentations that use the data and providing guidance on appropriate language and terminology.
Government agencies	Government agencies respect the rights of indigenous communities when sharing and communicating their data. This includes recognising and upholding the United Nations Declaration on the Rights of Indigenous Peoples, which affirms the right of Indigenous peoples to control their data and determine how it is used. Government agencies must take measures to protect the privacy and confidentiality of indigenous data. This includes implementing appropriate security measures, such as access controls and encryption, to ensure the data is not accessed or used inappropriately.
Academic institutions	Academic institutions must recognise and respect the sovereignty of indigenous communities when sharing and communicating their data. This includes acknowledging their right to control their data and determine how it is used. Academic institutions must develop policies that guide sharing and communication of indigenous data. These policies should be developed in consultation with indigenous communities and reflect their cultural protocols and values.
Data managers	Data managers respect the rights of indigenous communities when sharing and communicating their data. Data managers provide appropriate access and use agreements that reflect the rights and interests of the indigenous communities. These agreements should include provisions for data ownership, intellectual property rights, and the appropriate use and dissemination of the data. Data managers ensure that the data is of high quality and integrity and accurately reflects the indigenous communities' cultural and linguistic practices.
Data Users	Data users recognise and respect the sovereignty of indigenous communities when using their data. This includes acknowledging their right to control their data and determining how it is used. Data users follow appropriate protocols and guidelines for the use of indigenous data. This includes seeking approval from the indigenous community before using their data and complying with any agreed-upon data use restrictions. Data users acknowledge the source of the data and provide appropriate attribution to the indigenous community that provided it.

Data can have a longer lifespan than the research projects produced, allowing for continued analysis and reuse by other researchers even after the initial funding has ended (Owan & Bassey, 2019). However, it is important to note that the reuse of indigenous data must be done ethically and respectfully, fully considering the rights and interests of the indigenous peoples or communities involved. Table 8 provides a framework for indigenous data reuse in electronic archives.

PRINCIPLES AND VALUES GUIDING THE FRAMEWORK

The indigenous data management framework is guided by a set of principles and values that are important to indigenous communities. These principles and values reflect indigenous peoples' unique cultural and historical context and their relationship to data. Here are some of the principles and values that guide the framework:

- *Self-determination:* Indigenous communities have the right to determine how their data is collected, used, and shared. The framework respects and prioritises indigenous ownership and control over the data.

Table 8. A framework for indigenous data reuse in electronic archives

Stakeholders	Activity/Expectation
Research team	Researchers should obtain informed consent from the indigenous community or individuals who are the custodians of the data. Researchers should collaborate with indigenous communities and involve them in the research process. This may include co-designing research questions, co-analysing the data, and co-authoring research outputs. Researchers should acknowledge the sources and ownership of the data and ensure that appropriate credit is given to the indigenous community or individuals who have contributed to the data.
Indigenous community	Indigenous communities establish ownership and control of their data and have a say in how it is used and shared. Indigenous communities monitor who has access to their data and how it is used. This may involve setting up review committees or working with trusted intermediaries to manage access to the data. Indigenous communities should ensure that the reuse of their data in electronic archives benefits the community and is not exploitative.
Data managers	Data managers ensure that indigenous data is stored securely, and that confidentiality and privacy are maintained before reuse. Data managers implement appropriate metadata standards that accurately describe the indigenous data and ensure it can be easily discovered and accessed. Data managers implement appropriate data curation and preservation practices to ensure that the indigenous data is properly curated and preserved for future generations. Data managers ensure that the reuse of indigenous data in electronic archives complies with ethical and legal requirements, including informed consent, intellectual property rights, and data protection regulations.
Data Users	Data users respect indigenous communities' ownership and control of indigenous data. Data users acknowledge the source of the indigenous data and cite the original researchers or indigenous community from which the data was obtained. Data users should use indigenous data ethically and follow relevant guidelines and legal requirements. Data users should use the indigenous data responsibly to contribute to the well-being of the indigenous community. This may involve ensuring that the research benefits the indigenous community, respecting the cultural and spiritual significance of the data, and ensuring that the research is conducted in a culturally appropriate manner.

- *Indigenous knowledge:* Indigenous knowledge is valued and respected in the framework. The framework recognises that indigenous knowledge systems are diverse and complex and important for understanding the relationships between indigenous peoples, their environments, and their cultures.
- *Community engagement:* The framework is designed to engage indigenous communities in all stages of the data management process. Community members are involved in developing protocols and guidelines for data collection, processing, storage, retrieval, sharing, and reuse.
- *Cultural safety:* The framework promotes cultural safety by recognising indigenous peoples' unique cultural and historical context. The framework is designed to be sensitive to indigenous communities' cultural values and traditions and promote the well-being and safety of indigenous individuals and communities.
- *Respect for indigenous intellectual property rights:* The framework recognises the importance of indigenous intellectual property rights and seeks to protect them. Indigenous communities have the right to control their data, and the framework ensures that data is shared and reused in a way that respects these rights.
- *Trust and transparency:* The framework is designed to be transparent and trustworthy, with clear protocols and guidelines for all stages of the data management process. indigenous communities can trust that their data is being managed in a way that respects their rights and values.
- *Reciprocity:* The framework recognises that indigenous data management is two-way, with benefits flowing both ways. Indigenous communities may share their data with others, but they also have the right to benefit from using their data. The framework ensures that benefits are shared fairly and equitably.

EVALUATION METRICS FOR MEASURING THE SUCCESS OF THE FRAMEWORK

Several metrics can be used to evaluate the framework's success in managing indigenous data in electronic archives. These metrics include:

- ***Data quality*****:** The quality of the data collected, processed, and analysed can be evaluated based on criteria such as accuracy, completeness, and relevance.
- ***Data security:*** The security of the data can be evaluated based on criteria such as access controls, encryption, and backup and recovery procedures.
- ***Data accessibility:*** The accessibility of the data can be evaluated based on criteria such as ease of retrieval, availability of metadata, and compliance with relevant data-sharing policies and guidelines.
- ***Cultural and ethical considerations:*** The framework's success can also be evaluated based on criteria related to cultural and ethical considerations, such as the involvement of the indigenous community in the data collection process, respect for the privacy and dignity of individuals involved, and acknowledgement of the source and ownership of the data.
- ***User satisfaction***: The satisfaction of users, such as researchers and community members, with the framework, can be evaluated through surveys or interviews to assess ease of use, appropriateness, and effectiveness.
- ***Impact***: These impact of the framework on research, education, and community development can be evaluated by tracking the number of publications, presentations, and other outputs that use the data, as well as by assessing the benefits and outcomes of the research and other activities enabled by the framework.

CONCLUSION AND RECOMMENDATIONS

This chapter contributes to indigenous data management in electronic archives by proposing a comprehensive framework that reflects indigenous worldviews and practices. The framework provides an 8-step approach that includes key components such as community engagement, informed consent, and the development of culturally relevant metadata standards. The chapter concludes that indigenous data management in electronic archives offers valuable contributions to the empowerment of indigenous communities, ethical research practices, knowledge sharing, and reconciliation efforts. Indigenous data management in electronic archives has important implications for indigenous communities, research, and society. It empowers indigenous communities to own and control their data, allowing for preserving their cultural heritage and promoting self-determination. Researchers benefit from a better understanding of indigenous knowledge and practices, leading to more culturally sensitive and ethical research. Knowledge-sharing between indigenous and non-indigenous communities promotes social cohesion and ethical data management practices contribute to reconciliation efforts. Future research should focus on the ethical, legal, and social implications of indigenous data management, developing best practices and standards, and greater recognition and support within society. Future research should also focus on further exploring and refining these practices while addressing the broader societal implications and ensuring the recognition and support of indigenous data management within the field. Based on the framework developed in this chapter, it is recommended that iindigenous communities should participate in training programs to gain a comprehensive understanding of indigenous data management practices, ethical considerations and technical skills. Researchers and digital archivists should facilitate and conduct training programs, sharing expertise and knowledge on data management practices with indigenous communities. Indigenous communities should engage in collaborative governance processes by actively participating in the development and implementation of data management frameworks, policies, and decision-making.

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KEY TERMS AND DEFINITIONS

Cultural Heritage: Refers to the artefacts, practices, and traditions that are inherited from past generations and preserved as a cultural legacy. Cultural heritage includes tangible and intangible elements, such as artefacts, historical sites, language, music, and storytelling.

Electronic Archives: This refers to a collection of digital data, documents, and records preserved for long-term storage and access. Electronic archives can include various types of digital data, such as text, images, audio, and video.

Indigenous Data: This refers to the data that relates to or originates from Indigenous peoples, communities, cultures, or knowledge systems. This data often includes cultural, ecological, and linguistic information and is unique to Indigenous communities and their experiences.

Indigenous Research: Indigenous research is a type of research that is carried out by and with Indigenous peoples, communities, and knowledge systems. It emphasises the importance of Indigenous knowledge, perspectives, and values in shaping research questions, methodology, and outcomes.

Metadata: Data that provides information about other data. Metadata can include information about the context, structure, format, and content of digital data and is used to help organise and manage digital collections. In Indigenous data management, metadata can include information about data's cultural and historical significance and ownership and access.